Figure 3.9. String diagram for our DQN. The input is a 64-length Boolean vector, and the output is a 4-length real vector of Q values.

Figure 3.10. The loss plot for our first Q-learning algorithm, which is clearly down-trending over the training epochs.

Figure 3.11. The loss plot for Q-learning in random mode, which doesn't show any signs of convergence.

Figure 3.12. The idea of catastrophic forgetting is that when two game states are very similar and yet lead to very different outcomes, the Q function will get "confused" and won't be able to learn what to do. In this example, the catastrophic forgetting happens because the Q function learns from game 1 that moving right leads to a +1 reward, but in game 2, which looks very similar, it gets a reward of –1 after moving right. As a result, the algorithm forgets what it previously learned about game 1, resulting in essentially no significant learning at all.

Figure 3.13. This is the general overview of experience replay, a method for mitigating a major problem with online training algorithms: catastrophic forgetting. The idea is to employ mini-batching by storing past experiences and then using a random subset of these experiences to update the Q-network, rather than using just the single most recent experience.

Figure 3.14. The DQN loss plot after implementing experience replay, which shows a clearly down-trending loss, but it's still very noisy.

Figure 3.15. This is the general overview for Q-learning with a target network. It's a fairly straightforward extension of the normal Q-learning algorithm, except that you have a second Q-network called the target network whose predicted Q values are used to backpropagate through and train the main Q-network. The target network's parameters are not trained, but they are periodically synchronized with the Q-network's parameters. The idea is that using the target network's Q values to train the Q-network will improve the stability of the training.

Figure 3.16. The DQN loss plot after including a target network to stabilize training. This shows a much faster training convergence than without the target network, but it has noticeable spikes of error when the target network synchronizes with the main DQN.